Greener Under Pressure?

The Local Geography of Import Competition and Emissions in Swedish Manufacturing *

Albert Duodu
Aalto University, Department of Economics
albert.duodu@aalto.fi

Zouheir El-Sahli
International Monetary Fund
ZEl-Sahli@imf.org

Abstract

We examine how local import competition — measured across different spatial scales within Sweden — affects the CO_2 emission intensity of manufacturing firms. Using detailed data on the location of Swedish manufacturing firms, we show that greater local import competition lowers the emission intensity of upstream domestic suppliers, with the effect diminishing as the distance between producers and importers increases — spatial variation that national measures of import competition cannot detect. We trace this decline to a single firm response. Facing localized competition, suppliers raise productivity and cut marginal costs (a technique effect) and escape the contested segment by upgrading quality and reallocating toward core products while shedding peripheral, emission-intensive ones (a composition effect). Because escaping competition through differentiation raises prices and markups even as marginal costs fall, the often-puzzling combination of procompetitive pressure and rising markups is a signature of quality upgrading rather than weaker competition. We also find that part of the adjustment is the offshoring of carbon-intensive inputs, so the decline in domestic emission intensity overstates the global emission saving.

Keywords: Import competition, trade and environment, emissions, heterogeneous firms

JEL: F14, F18, Q56, Q58, R13

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1 Introduction

International trade reshapes domestic production networks. As firms increasingly rely on imported intermediate inputs and participate in global value chains (GVCs), trade shocks alter competitive pressures not only for direct importers but also for their domestic suppliers. These adjustments have potentially important environmental consequences. Yet while a large literature studies how trade affects productivity, markups, and labor outcomes, much less is known about how trade-induced competitive pressure influences firms' environmental performance—particularly through indirect supplier linkages.

When a domestic manufacturer begins importing a product that was previously sourced locally, upstream domestic producers face intensified import competition (henceforth IC). In response, these firms may contract, exit, upgrade, or reorganize production. Each of these adjustments can alter emission intensity. On the one hand, competitive pressure may induce efficiency improvements, product upgrading, and adoption of cleaner technologies, reducing emissions per unit of output (Newman et al., 2023; Jabbour et al., 2019; Copeland et al., 2022). On the other hand, if firms operate under increasing returns to scale or face adjustment frictions, IC could reduce output scale in ways that increase emission intensity (Shapiro and Walker, 2018; Cherniwchan et al., 2017). Whether firms become 'greener under pressure' is therefore theoretically ambiguous and ultimately an empirical question.

In Sweden, domestic upstream suppliers represent a non-trivial share (about 59%) of total manufacturing emissions. While much of the trade–environment literature focuses on the direct environmental response of importing firms, production networks imply that competitive shocks propagate upstream through geographically concentrated supplier relationships. If supplier adjustments are environmentally significant, analyses that consider only importing firms may underestimate the aggregate environmental impact of trade shocks. In this paper, we quantify the contribution of upstream suppliers to overall manufacturing emissions and show that localized competitive pressures account for a meaningful share of observed emission reductions.

IC is usually measured at the national level with the assumption that competition increases symmetrically within an economy as soon as an imported product crosses the border (Lehr, 2025; Autor et al., 2013; Gullstrand and Knutsson, 2019). This approach underlies much of the trade–environment literature, including studies of pollution and energy efficiency responses to IC (Cherniwchan et al., 2017; Gutiérrez and Teshima, 2018; Liu et al., 2023). However, buyer–seller relationships are geographically concentrated (see e.g., Bernard et al., 2019; Hillberry and Hummels, 2008), and production networks exhibit strong spatial frictions (see e.g., Syverson, 2007). If supplier linkages are locally clustered, competitive pressure from imports is unlikely to diffuse uniformly across space. Instead, exposure should decay with distance and concentrate within local production networks. In such a setting, national measures may misstate effective competition—particularly for outcomes that depend on production technologies, supply chains and local market structure, such as CO_2 emission intensity (Shapiro, 2021).

To address these issues, we build on the spatial import-competition framework developed by

Gullstrand and Knutsson (2019), who show that IC in Sweden operates through geographically localized buyer–seller networks. Extending their approach, we use detailed spatial data on firm locations to construct firm-level exposure to localized IC shocks. Because direct buyer–seller linkages are not observed, we model upstream exposure based on the proximity between importing firms and domestic producers of the same products, allowing competitive pressure to decay with distance. Our contribution is not methodological. Rather, we extend this spatial framework to a new environmental dimension by examining how localized IC affects upstream firms' emission intensity.[1] While previous work has documented productivity gains and product reallocation in response to IC, the environmental consequences of these adjustments—particularly among indirectly affected domestic suppliers—remain largely unexplored. By quantifying these indirect environmental responses, we provide a more complete assessment of how trade shocks reshape domestic emissions.

In this paper, we combine administrative data on Swedish manufacturing firms' production, trade flows, and energy use for the period 2005–2014. We construct firm-level exposure to localized IC by linking product-level import flows to the geographic proximity between importing firms and domestic producers of the same goods. This generates variation in competitive pressure across firms, space, products, and time. A central empirical challenge is that domestic demand conditions may simultaneously affect both imports and firms' environmental performance. To address this concern, we implement a shift-share instrumental variable strategy that exploits plausibly exogenous variation in world export supply across products and source countries. The instrument interacts firms' pre-sample exposure shares with global supply shocks, thereby isolating the component of local IC driven by foreign export growth rather than domestic conditions.

This paper makes three main contributions. First, we advance the trade–environment literature (see Copeland et al., 2022 for a review) by showing that the environmental consequences of IC extend beyond directly importing firms and propagate through geographically localized supplier networks. While prior research documents how IC shapes productivity, markups, and product mix, and more recent work examines environmental responses among importing firms (Akerman et al., 2024; Leisner et al., 2023; Liu et al., 2023), much less is known about whether and how trade-induced competitive pressure affects the emissions of upstream domestic producers. By isolating localized competitive pressure while controlling for firms' own imports, we identify an indirect transmission channel operating through nearby importers within the same supply network. We demonstrate that this indirect channel is both economically meaningful and spatially heterogeneous. IC measured at fine geographic scales induces significant reductions in emission intensity among local suppliers and these effects decay with distance.

[1] To further explain this, assume a Swedish car manufacturer (downstream Swedish firm) that outsources glass for its cars. This car manufacturer can purchase glass from a (nearby) local producer, i.e. the Swedish upstream firm, or a foreign manufacturer. If the Swedish downstream car manufacturer decides to import the glass from a foreign producer, this represents IC for the Swedish glass manufacturer. How the latter firm responds to this competitive pressure is likely to affect its environmental performance, and specifically its emission intensity.

By uncovering this upstream transmission mechanism, we show that trade shocks reshape environmental performance throughout localized production networks, rather than solely within importing firms.

Second, we contribute to the growing literature on the spatial organization of production networks (Gullstrand and Knutsson, 2019, Bellone et al., 2016, Bernard et al., 2019, Hillberry and Hummels, 2008, Autor et al., 2013) by demonstrating that the environmental consequences of trade shocks are fundamentally distance-dependent. Building on Gullstrand and Knutsson (2019), we show that the emission response to IC declines sharply with geographic distance between importers and domestic suppliers. This spatial attenuation implies that the environmental impact of trade is highly localized: competitive pressures transmit most strongly within clustered production networks and weaken as linkages stretch across space. National measures, by averaging exposure, blur this localized pressure and the supplier adjustments it triggers; the effect is concentrated among the most exposed firms rather than spread uniformly across the country. In emphasizing geography, we connect two literatures that have largely developed in parallel—the spatial organization of production (Bernard et al., 2019; Hillberry and Hummels, 2008) and spatial environmental economics (Balboni and Shapiro, 2025; Copeland et al., 2022; Felbermayr et al., 2025)—by showing that where production is located shapes how trade shocks translate into emissions.

Third, we provide new evidence on the mechanisms through which trade-induced competitive pressure reshapes the firms' environmental performance (Newman et al., 2023; Shapiro and Walker, 2018; Dussaux et al., 2023; Akerman et al., 2024; Leisner et al., 2023; Jabbour et al., 2019; Antoniades, 2015). While prior work shows that trade exposure reduces emission intensity primarily through within-firm productivity gains—what Shapiro and Walker (2018) term a technique effect—and importer-level technological upgrading (Akerman et al., 2024), these analyses focus on firms directly engaged in trade/importing and find limited scope for pollution haven dynamics. We show that once IC is measured at fine geographic scales and traced through local production networks, the adjustment is broader and more complex.

Our findings suggest that increased local IC significantly reduces the firm's CO_2 emission intensity, and that this effect is substantially stronger when exposure is measured at fine geographic scales rather than at the national level, indicating that the environmental response to trade is mediated by local production networks. We present a simple conceptual framework (see section 3) that organizes the mechanism and reconciles the results that, if taken separately, may appear inconsistent. Facing localized competition, upstream domestic suppliers see their productivities increase and their marginal costs diminish (a technique effect), and they respond by upgrading quality and pivoting toward core products while shedding peripheral, emission-intensive ones (a composition effect). Because responding to competition through differentiation raises prices and markups even as marginal costs fall, the combination of pro-competitive pressure and rising markups reflects quality upgrading. Strikingly, we further detect partial reallocation of more carbon-intensive production stages abroad—an offshoring margin that is not apparent in importer-focused analyses using the same Swedish context. By jointly quanti-

fying productivity improvements, product-mix adjustments, abatement investment, and carbon offshoring among upstream domestic suppliers—many of which are not direct importers—we uncover a broader reorganization of production in response to trade shocks. This evidence suggests that competitive pressures reshape environmental performance not only within importing firms, but throughout localized supply networks, revealing margins of adjustment that are obscured when attention is confined to importers alone.

This paper proceeds as follows. Section 2 describes the data, sample construction, and measures of IC. Section 3 specifies the empirical strategy as well as the instruments. Section 4 provides the results and various robustness checks. Section 5 shows the channels at work and section 6 concludes.

2 Data and descriptive statistics

2.1 Firm-level Emissions and Abatement

The data used in this paper originate from Statistics Sweden (SCB) and consist of several merged data sets covering Swedish manufacturing firms with ten or more employees. The first database is the energy consumption database, which contains information on the energy consumption of all manufacturing plants. Firms report detailed information about the annual consumption of different fuel types with their units (e.g., liters of diesel, mwH of electricity, m^3 of natural gas etc.). CO_2 emissions can be calculated by using the fuel-specific CO_2 emission coefficients provided by SCB. Hence, CO_2 emissions are accurately calculated from fuel inputs. Our dependent variable will be emission intensity measured as emissions per Swedish Krona (SEK) of value-added.[2] Table B.1 presents descriptive statistics for the emissions and emission intensities of firms. On average, firms emit approximately 45,000 tons of CO_2 annually, translating to about 0.017 tons of CO_2 per SEK value added. The median values of emissions and emission intensity are consistently lower than the averages, indicating right skewness and a high concentration of emissions among a small number of firms.

Abatement stands for the activities by the firm to reduce pollution. We obtain annual abatement data from a survey conducted by SCB where firms report their abatement investments and expenditures (measured in thousands SEK). Specifically, firms are asked to report any investments and expenditures they have made in machinery and equipment aimed at reducing pollution and emissions, as well as any expenses associated with investing in cleaner machines and technology.[3] The abatement data come from a semi-random sample of manufacturing

[2]We also consider alternative measures of emission intensity calculated as emissions per unit output (see e.g., Barrows and Ollivier, 2021) and value of output (see e.g., Shapiro and Walker, 2018). All results in this paper are robust to using the alternative measures.

[3]SCB defines pollution abatement as capital expenditures for methods, technologies, processes, or equipment designed to collect and remove pollution after its creation, prevent its spread, and treat and dispose of pollutants generated by the company's operations. These include protection of ambient air and climate, waste(water) management, protection of soil and groundwater, among others. Some of these technologies such as thermal oxidizers and catalytic converters improve combustion efficiency and optimize fuel usage, which can indirectly reduce total CO_2 emissions from auxiliary fuel consump-

firms, which includes all manufacturing firms with more than 250 employees, 50% of firms with 100–249 employees, and 20% of firms with 50–99 employees. Figure B.1 shows the trends in abatement expenditure and investment. We observe a notable increase in average abatement investment and expenditure after 2008. Nevertheless, the average abatement expenditure has seen a slight decline since 2010, hinting at a potential trade-off between expenditure and investment. This suggests that firms' commitments and allocation of funds for environmental expenditure and investment may vary.

2.2 Production, Trade, and Geography

On the production side, we merge several administrative databases obtained from SCB. First, we use the Production of Commodities and Industrial Services (IVP) database, which provides annual information on quantities and values of production at the 8-digit product level (which we aggregate to the 6-digit equivalent as we will explain later) for approximately 6,200 Swedish manufacturing firms. The IVP covers private manufacturing firms employing at least 20 workers. Because geographic plant identifiers are incomplete prior to 2005, we restrict the analysis to the period 2005–2014, ensuring consistent firm-plant linkage throughout the sample.

Second, we obtain trade data from the International Trade in Goods (ITG) database. This database provides detailed country and product trade information for all Swedish firms, encompassing values and quantities of imports and exports at the 8-digit product level. ITG records the universe of Swedish firms' imports and exports, including transaction-level values and quantities at the 8-digit product level and by partner country. Production and trade are reported at the eight-digit Combined Nomenclature (CN8) level. Because the world export supply underlying our instrument is observed only at the six-digit Harmonized System (HS6) level in UN COMTRADE, we aggregate all product information to the HS6 heading, indexed by $p \in \mathcal{P}$. Aggregating to HS6 also ensures that near-identical varieties carrying distinct CN8 codes are treated as a single competitive product, so competition among close substitutes is not mechanically missed. We also restrict attention to products that are domestically produced according to the IVP production register. This restriction is economically and econometrically important. It ensures that our measure of IC is defined over products with an active domestic counterpart, thereby capturing genuine competitive pressure rather than mechanical import exposure. It also mitigates attenuation bias arising from imports of goods that have no domestic production analogue..

Third, we draw from the Structural Business Statistics (FEK) database, which provides comprehensive firm-level information on employment, industry classification, sales, assets, investment, and other balance-sheet variables. FEK allows us to measure firm performance and

tion. We note that most of these are end-of-pipe technologies aimed primarily at conventional pollutants (e.g., SO_x, NO_x, particulates); since CO_2 has no comparable end-of-pipe abatement technology short of carbon capture, we interpret reported abatement spending as a proxy for the firm's broader environmental effort—energy-efficiency upgrades, cleaner-technology adoption, and fuel switching—rather than as direct CO_2 abatement capital. Source: https://www.epa.gov/air-emissions-monitoring-knowledge-base/monitoring-control-technique-thermal-oxidizer

control flexibly for observable firm characteristics.

Together, the IVP, ITG, and FEK databases form a comprehensive, transaction-level panel that links detailed product-level trade exposure to firm-level production and financial outcomes within a unified administrative framework. Of particular relevance to this paper, the data includes the location coordinates (latitudes and longitudes) of plants and firms within Sweden. This enables us to geo-locate all firms, producers and importers. In part of the spatial analysis, we use a fine geographical division of Sweden called SAMS (Small Areas for Market Statistics). SAMS cells are defined by SCB as the smallest functional economic units that capture market integration. SAMS divides Sweden's 290 municipalities into 9209 small areas, which allows us to create spatial variables without worrying about the unevenness of municipal borders. Each cell's centroid and coordinates are obtained from the SCB's GIS register. IC is *local* in the sense that we measure it at the SAMS level—the imports of other manufacturing producers located in the same or nearby SAMS cells—rather than at the industry or regional level. Note that as robustness, we will also use direct distance between firms - the concept of nearest neighbors - in our analysis.

Although customs records identify the importing firm, they do not report the physical location where imports are used. This presents a central measurement challenge: many firms operate multiple plants, and the geographic incidence of imported inputs or final goods may differ across establishments. To recover a credible measure of local import exposure, we merge firm-level trade records with establishment-level information from the Central Business Register (CFAR), which provides annual plant identifiers, employment, and precise geographic location (SAMS). Subsequently, we define plant-level imports as follows. Let $\mathcal{J}(f,t)$ denote the set of plants j owned by firm f in year t, and let L_{fjt} denote plant employment. Firm-level imports are allocated to plants using an employment-share rule that is deliberately conservative and transparent:

$$\tilde{M}_{fjpot} \;=\; M_{fpot} \times \omega_{fjt}, \qquad \omega_{fjt} \;=\; \frac{L_{fjt}+1}{\sum_{j' \in \mathcal{J}(f,t)} (L_{fj't}+1)}. \tag{1}$$

In the above equation, $\tilde{M}_{fjpot}$ stands for plant-level imports of firm f in product p from country o in any given period t. The $+1$ adjustment prevents undefined shares when employment is missing or zero for small units and ensures all multi-plant firms are allocated strictly positive weights. Equation (1) yields plant-level imports $\tilde{M}_{fjpot}$ that respect firm totals by construction and place the geography of imports where the firm's economic activity is plausibly located. Importantly, the allocation step anchors trade exposure in the firm's physical production footprint, rather than attributing all imports to headquarters or to an arbitrary location. This is crucial for a design that seeks to measure localized competitive pressure.

On the production side, emissions are recorded at the firm level rather than the plant level. Because IC varies geographically across plants, aggregating outcomes for multi-plant firms would introduce spatial misclassification. A multi-plant firm operating in heterogeneous local markets would be assigned an exposure equal to a weighted average of distinct plant-level shocks, while its observed emissions reflect the sum of potentially heterogeneous plant

responses. This aggregation induces classical measurement error in exposure and mechanically attenuates estimates of local competitive effects. To eliminate this within-firm spatial ambiguity, our baseline estimating sample focuses on single-plant producing firms. For these firms, the geographic unit of exposure coincides exactly with the unit at which production and emissions are observed, ensuring a clean mapping between local IC and firm outcomes. Importantly, this restriction does not mechanically confine the analysis to small or peripheral producers: single-plant firms account for approximately 70% of firms in the production database and roughly 59% of total domestic manufacturing emissions.

We further restrict the sample to active firms with energy and emissions information and positive sales, excluding firms with extended periods of inactivity. In addition, the analysis focuses on intermediate goods producers, who plausibly compete with imported intermediate inputs in local markets. Notably, manufacturing firms account for approximately 80% of total imports in the compiled sample, reinforcing the relevance of IC within the manufacturing sector.

2.3 Measuring Import Competition

In this section, we present how we measure IC exposure faced by the upstream firm producing an intermediate input product that is simultaneously imported by the downstream manufacturer. The construction is designed to capture a conceptually precise object: *the intensity of foreign competitive pressure in the local market environment surrounding the firm*, allowing for spatial decay and market interaction across neighboring locations.

SAMS geography and the spatial structure of competition. We measure IC at the level of Small Areas for Market Statistics (SAMS), a high-resolution geographic partition designed to approximate economically meaningful neighborhoods. Let $s \in \mathcal{S}$ be the set of all SAMS areas, with individual SAMS areas indexed by $s \in \mathcal{S}$, and let each plant j be located in $s(j)$ with centroid coordinates (x_s, y_s). The fine spatial resolution is essential: competitive pressure in manufacturing operates through transportation networks, supplier linkages, and local labor markets that do not conform neatly to broader administrative boundaries.

We begin by aggregating allocated plant imports to the SAMS–product–year level. Let f index importing firms. Total imports of product p in SAMS s and year t are defined as

$$M_{spt} = \sum_{h} \sum_{j \in \mathcal{J}(f,t):\, s(j)=s} \sum_{o} \tilde{M}_{fjpot}, \tag{2}$$

where $\tilde{M}_{fjpot}$ denotes plant-level imports defined in equation (1). Because the allocation preserves firm totals exactly, M_{spt} reflects the spatial distribution of observed customs imports without altering aggregate trade volumes. This step matters: competitive exposure is inherently geographic, and attributing all imports to a firm's headquarters would mechanically mismeasure the spatial incidence of competition.

A key identification concern is that a producer's own imports should not mechanically inflate its measured exposure to IC. If a firm that imports more were mechanically assigned

greater competitive pressure, the exposure measure would partly reflect endogenous sourcing decisions rather than external competition. To eliminate this reflection problem, we explicitly remove the producing firm's own imports from the local market environment.

Let i index producing firms and let $s(i,t)$ denote firm i's SAMS location in year t. Define

$$M_{spt}^{(-i)} = M_{spt} - \sum_{j \in \mathcal{J}(i,t): s(j)=s} \sum_{o} \tilde{M}_{ijpot},$$

that is, total imports in SAMS s excluding imports attributable to firm f itself.

Localized import competition. Competitive pressure is not confined within administrative borders. Producers plausibly compete with importers located in nearby markets, with competitive intensity declining in geographic distance. We therefore model exposure as a spatially weighted average of surrounding import activity. Let W denote a spatial weight matrix over SAMS satisfying $w_{ss'} \geq 0$, $w_{ss} = 0$, and row normalization $\sum_{s' \neq s} w_{ss'} = 1$, $w_{ss'}$ being the ss^{th} element of W[4]. Our baseline specification uses inverse-distance weights,

$$w_{ss'} \propto d_{ss'}^{-1},$$

where $d_{ss'}$ denotes the geographic distance between SAMS centroids. This structure allows competitive influence to decay smoothly with distance rather than discontinuously at arbitrary borders.

We first define the external market component, which captures import intensity in neighboring areas. This can be written as:

$$\mathcal{IC}_{spt}(W) = \sum_{s' \neq s} w_{ss'} M_{s'pt}, \tag{3}$$

Localized IC facing firm f in product p and year t can therefore be written as:

$$\mathcal{IC}_{ipt} = \underbrace{\sum_{s' \neq s(i,t)} w_{s(i,t)s'} M_{s'pt}}_{\text{External markets}} + \underbrace{M_{s(i,t)pt}^{(-i)}}_{\text{Same-area competitors (excluding own)}}. \tag{4}$$

In equation (4), the first term measures competitive pressure arising from nearby SAMS areas, also defined in equation (3). The second term captures competing imports within the same local area, net of the firm's own imports. Together, these components yield a measure of external competitive exposure that is geographically continuous and free from mechanical self-

[4]Row-normalization implies that $\mathcal{IC}_{spt}(W)$ measures the distance-weighted *average* import intensity in surrounding areas, rather than the total import mass within reach. This is a deliberate modeling choice: without row-normalization, firms located in regions surrounded by many SAMS areas (typically urban centers) would mechanically record higher IC than firms in sparsely populated regions, even if the average competitive intensity is the same. Row-normalization ensures comparability across SAMS areas of heterogeneous size and neighbor density. We verify that results are robust to unnormalized inverse-distance weights, which scale the measure by total surrounding import mass; the estimates are qualitatively similar and available upon request.

influence. The decomposition clarifies that competition operates both within narrowly defined markets and across adjacent regions.

To assess the spatial reach of competition, we additionally construct band-limited weight matrices isolating (i) 0–500 km and (ii) 500–1000 km neighbors, each row-normalized within band. These alternative weight structures allow us to decompose competitive pressure into near-market and mid-range components and assess how rapidly competitive effects decay with distance.

From product-level exposure to firm-level competition. Finally, we aggregate product-level exposure to the firm level using predetermined product weights. Let θ_{ip0} denote firm i's baseline production (or input-use) share in product p, measured prior to the main sample window and held fixed over time. Firm-level IC is defined as

$$\mathcal{IC}_{it}^{SAMS} = \sum_{p \in \mathcal{P}} \theta_{ip0} \, \mathcal{IC}_{ipt}. \tag{5}$$

This construction yields a firm-year measure of IC that is (i) granular at the product level, (ii) localized through SAMS geography and spatial spillovers, and (iii) predetermined in composition through baseline product weights. The construction ensures that variation in exposure reflects external shifts in competitive pressure rather than endogenous adjustments in product mix or importing behavior.

2.4 Data Descriptive

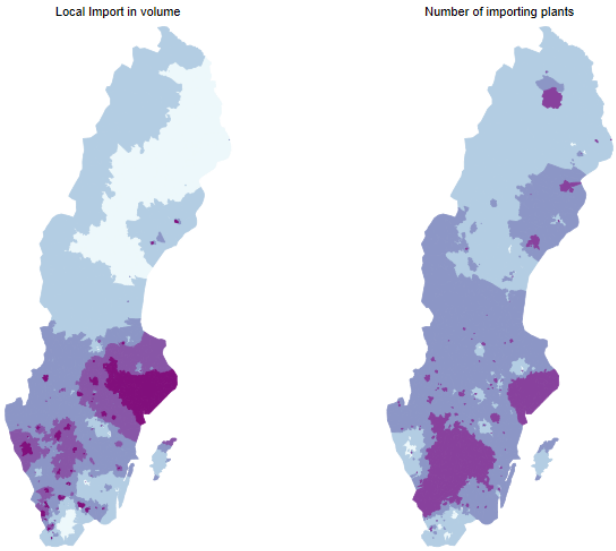
Table B.2 also presents a summary of production data and other firm-related information, including employment, markup, and sales. After restricting our sample to active firms operating a single plant and selling intermediate products locally, we end up with 2,279 manufacturing firms. However, within this sample, we have emission information available for only 1,497 firms (approximately 65% of the sample is retained). On average, the producing firm in our sample produces about 3 products with a maximum number of 68 products, and average sales of 930 million SEK. The average firm's sales from the core (main) product as a share of revenue is about 53% - a product is classified as a core product when it accrues more than 50% of the entire firm revenue. Moreover, the average share of products in the firm's total production exposed to IC pre-sample is about 67%. On average, about 59% of the firms' products are classified as a core product, whereas the rest are fringe products. We calculate the markup for both single-product and multi-product firms using techniques initially proposed by Hall et al. (1986) and further refined by De Loecker et al. (2016).[5]

Moreover, we present descriptive statistics about imports and the geographic location of importers vis-à-vis the local producers in Table B.3. The average import value for manufacturing firms across Sweden is around 190 million SEK, while at the SAMS level, it hovers around 9

[5]We obtain the markups for individual products produced by firms by dividing the output elasticity of materials by the materials share of total revenue.

million SEK, suggesting large concentrations of imports in some SAMS areas. Regarding the location of the producing firm relative to the nearest importing firms, we note that the average import value is higher for the first nearest importing firm compared to the subsequent ones. This is a first indication that IC is on average more intense in the vicinity of the firm. Analyzing the distance between importers and producers, we find that the average distance from the first nearest importing neighbor is approximately 23.33 km, while the third nearest is almost twice that distance at 66.32 km. These distances have larger coverage than the average SAMS area, with a radius of around 2 km, and the average municipality, with an average radius of 18 km. Figure B.2 displays the distribution of producers and importers across SAMS regions. The distribution is skewed with most SAMS regions having fewer producers and importers, and a few having many.

Figure 1: Spatial distribution of import exposure



Note: The shading demarcates distinct regions into quintiles, with the darkest shade symbolizing the highest quintile and the faintest shade signifying the lowest quintile. All figures incorporate spatial lags, employing weight adjustments standardized by rows, which are determined by distance.

Figure 1 displays the spatial distribution of IC and the share of importing plants in the total local (SAMS) number of manufacturing firms. In the left panel of the graph, we compute the import volume and number of importing plants for each SAMS area plus their spatial lags (with the spatial-weights based on distance). We observe a notable concentration of raw import volume along a corridor extending from Stockholm, reflecting the high degree of economic activity in these areas. In the right panel, we observe that the numbers of importing plants tend to cluster around Stockholm and in the central part of Southern Sweden.

3 Conceptual Framework

This section develops the economic intuition linking localized import competition to firms' environmental performance. The framework identifies the main adjustment margins available to upstream domestic suppliers and derives a set of testable predictions regarding emission intensity, productivity, pricing, markups, product reallocation, abatement, and offshoring. These predictions provide the roadmap for the empirical analysis that follows.

3.1 Emission intensity: scale, technique, and composition

First, we follow the decomposition that organizes the modern trade-and-environment literature (Copeland and Taylor, 2004; Shapiro and Walker, 2018; Cherniwchan et al., 2017). A firm's CO_2 emissions can be written as the product of a *scale* term (how much it produces), a *technique* term (emissions per unit of output, given its product mix), and a *composition* term (which products it makes):

$$\underbrace{E_{it}}_{\text{total CO}_2} = \underbrace{Y_{it}}_{\text{scale}} \times \underbrace{\sum_{p \in \mathcal{P}_{it}} s_{ipt}\, \theta_{ipt}}_{\text{intensity = technique} \times \text{composition}}, \tag{6}$$

where Y_{it} is real output, s_{ipt} is the share of product p in the firm's output, and θ_{ipt} is the emission intensity of product p. Our outcome is the bracketed term, *emission intensity* $EI_{it} \equiv E_{it}/Y_{it}$ (measured per unit of value added). Two implications follow immediately and discipline the interpretation throughout.

First, intensity falls either because each product is made more cleanly (the technique effect, a change in θ_{ipt}) or because the firm shifts its mix toward cleaner products (the composition effect, a change in s_{ipt}). Our mechanism analysis is organized around exactly this distinction.

Second, a fall in intensity does *not* by itself imply a fall in total emissions. If competitive pressure also raises scale Y_{it}, total CO_2 can move in either direction, and only total CO_2 is relevant for climate. We therefore treat scale as a distinct object: we report the effect of import competition on output and value added.

3.2 How a supplier responds to localized import competition

The unit exposed to competition in our setting is the *upstream domestic supplier*, not the firm that imports. When nearby downstream plants substitute toward foreign varieties of an input, the domestic suppliers of that input face a contraction in residual demand. Following the heterogeneous-firm tradition (Melitz, 2003; Melitz and Ottaviano, 2008), this shock has two margins of adjustment that operate together.

The technique margin: lower marginal cost. A negative demand shock raises the effective return to cost reduction. Surviving suppliers cut slack, raise productivity, and reallocate toward more efficient uses of inputs (Newman et al., 2023; Aghion et al., 2005). Because energy is an input, higher productivity mechanically lowers fuel use per unit of output: the technique term

in (6) falls. This is the standard pro-competitive efficiency channel, and on its own it predicts both lower marginal cost and lower emission intensity.

The differentiation margin: The pro-competitive efficiency channel does not, by itself, predict the sign of markups. The naive intuition—that competition compresses markups—holds only *conditional on a fixed product and fixed quality*. It does not describe a firm that responds to competition by moving away from head-to-head rivalry. Under variable markups and endogenous quality (Melitz and Ottaviano, 2008; Antoniades, 2015), a supplier facing a commoditized, import-contested segment optimally *escapes competition* (Aghion et al., 2005) by upgrading quality and reallocating toward its more differentiated, higher-margin varieties. This differentiation raises the firm's price and its markup even as its marginal cost falls, because the firm is no longer selling the same good into the same contested market—it is selling a repositioned, higher-quality product over which it retains pricing power. If firms respond to localized import competition by upgrading quality and escaping direct price competition, rising prices and markups become a natural prediction despite stronger competitive pressure. Observing such a pattern would therefore be consistent with an escape-through-differentiation mechanism rather than weaker competition.

The same differentiation logic links directly back to emissions through the composition term in (6). Escaping competition means concentrating on core products and shedding peripheral lines. In our data—as in the broader quality-and-trade literature—peripheral, low-margin products are disproportionately the firm's dirtiest (Antoniades, 2015; Shapiro, 2021). Dropping them, and upgrading the survivors, lowers emission intensity through composition at the same time that it raises the average markup. The markup result and the emission result are thus two readings of one adjustment.

Reconciling large markup gains with small measured TFP gains. A natural objection is that markups rise substantially while measured TFP rises only modestly. This wedge is expected, not anomalous. Our productivity measure is revenue-based (it is recovered from deflated sales), so when a firm upgrades quality and raises prices, part of the true efficiency and quality gain is absorbed into the price index rather than into measured physical productivity (De Loecker et al., 2016). The De Loecker–Warzynski markup, by contrast, captures the full price–marginal-cost wedge, quality included. The gap between a large markup response and a small revenue-TFP response is therefore exactly the fingerprint of quality upgrading. We test this interpretation directly below by examining a price-inclusive productivity measure (value added per worker) and by asking whether the price response is compositional rather than within-product.

Offshoring and leakage. Finally, repositioning toward cleaner core production can be achieved partly by *relocating* the dirtiest stages abroad rather than abating them at home. A supplier may import the carbon-intensive inputs it once produced in house, which lowers domestic emission

intensity through composition but raises emissions embodied in its imports. This is a leakage margin, and we measure it explicitly so that the domestic intensity reduction is not mistaken for a global one.

3.3 Empirical implications

The framework above organizes the empirical analysis that follows. Rather than viewing the various outcomes as separate exercises, we examine whether they are consistent with a common adjustment process triggered by localized import competition.

We begin by asking whether competitive pressure improves firms' environmental performance by reducing emission intensity. We then examine whether any reduction reflects a *technique effect*—cleaner production through higher productivity and lower marginal costs—or a *composition effect* arising from changes in the firm's product mix. Next, we investigate whether suppliers respond by escaping direct competition through product differentiation, which predicts higher prices and markups despite stronger competitive pressure. We then examine whether firms concentrate production on cleaner core products while discontinuing more emission-intensive peripheral products, and finally assess whether part of the adjustment occurs through increased reliance on imported carbon-intensive inputs, implying that improvements in domestic emissions may partly reflect the relocation of emissions abroad.

Overall, these empirical tests allow us to distinguish among the different channels through which localized import competition may affect the environmental performance of upstream domestic suppliers. The results are presented sequentially in the following sections.

4 Empirical Approach

To estimate the effect of localized IC exposure on the CO_2 emission intensities of producers, we consider the following specification:

$$EI_{it} = \beta_0 IC_{it}^{SAMS} + \theta_i + \theta_{bt} + \theta_{mt} + \epsilon_{it}, \tag{7}$$

where EI_{it} is the amount of emissions per SEK of value-added (in logs) of producing firm i in year t and IC_{it}^{SAMS} captures the IC (in logs) faced by producing firm i at time t. We condition import exposure on firm fixed effects (θ_i) to control for individual firm unobserved heterogeneity. Additionally, we control for industry-year fixed effects (θ_{bt}), with the industries being defined at the 3-digit industry (SNI) codes, to capture broad changes in demand, production, industry emission intensity levels, and imports by wholesalers of different goods. We further include municipality-year fixed effect (θ_{mt}) to control for shocks at the municipal level in labor markets as well as changes in environmental regulations, which are set at the municipal level in Sweden. With municipality×year fixed effects, identification comes entirely from *within* municipality-year variation: two single-plant producers in the same municipality and year are compared when they face different localized IC, which arises because they produce different

products and sit in different SAMS cells at different distances from importing plants. The fixed effects thus absorb any municipality-wide demand, policy, or labor-market shock, leaving firm-by-product-by-location variation in IC as the identifying variation.

Our measure of IC, which largely depends on import demand of Swedish importers, may suffer from endogeneity issues. For example, firms may be encouraged to import intermediate goods if they are located close to low-productivity producers. Hence, changes in supply and demand conditions may feed into each other, which would lead to reverse causality. Other sources of endogeneity are omitted variable bias and measurement error. To mitigate these concerns and estimate the causal effects of IC exposure, we must isolate the supply-driven increase in imports (components that are caused by arguably exogenous increases in trade). Following Hummels et al. (2014) and recent discussions by Goldsmith-Pinkham et al. (2020), Borusyak et al. (2022a) and Borusyak et al. (2022b), we use global supply shocks directed to countries other than Sweden and its neighboring countries as instruments.

4.1 Shift–Share Identification

We obtain world export supply from UN COMTRADE at the HS6 level—the same level at which import competition is measured—so no cross-level aggregation is required. Because the HS nomenclature was revised twice during our sample period (in 2007 and 2012), we map every vintage to a single fixed classification using official concordances before constructing the shocks, ensuring that each product is defined consistently over time (Van Beveren et al., 2012; Pierce and Schott, 2012). For each product p and year t, we construct

$$WX_{pct} = \log\left(\text{ExportSupply}_{pt}^{(-SE)}\right),$$

where the world export supply excludes flows to Sweden and its neighboring countries (Denmark, Finland, Germany, and Norway). These shocks capture changes in global supply conditions — such as productivity growth, trade liberalization, or technological upgrading in exporting countries — that affect the relative availability and price of product p independently of Swedish demand.

To translate global shocks into local competitive pressure, we follow the standard shift–share logic. For each importing firm f, define the pre-sample import share of product p as

$$\phi_{fp0} = \frac{M_{fp0}}{\sum_{p' \in P} M_{fp'0}},$$

where shares are measured prior to the estimation window and held fixed throughout. These predetermined shares capture firms' baseline exposure to each product category.

We then aggregate export shocks across importers using the same spatial structure that defines localized IC in equation (4). Let Ω_s denote importers located in SAMS s, and let $w_{ss'}$

denote inverse-distance weights. The product-level shock to location s is

$$Z_{spt} = \sum_{s' \neq s} w_{ss'} \sum_{f \in \Omega_{s'}} \phi_{fp0} WX_{pt} + \sum_{f \in \Omega_s} \phi_{fp0} WX_{pt}.$$

This expression mirrors the structure of localized IC: global product shocks are transmitted through geographically proximate importers, with competitive influence decaying smoothly in space.[6]

For a producing firm i located in $s(i)$, the product-level instrument is $Z_{ipt} = Z_{s(i),pt}$. Aggregating across products using the firm's predetermined production weights θ_{ip0} yields the firm-level instrument

$$IV_{it}^{SAMS} = \sum_{p \in P} \theta_{ip0} Z_{ipt}.$$

The instrument therefore varies at the firm-year level through three sources: (i) exogenous product-level world export shocks, (ii) cross-sectional differences in baseline import exposure of geographically proximate importers, and (iii) predetermined product weights of the producing firm. Because world export growth is measured outside Sweden and shares are fixed prior to the sample period, the identifying variation is driven by foreign supply shifts rather than domestic demand or contemporaneous firm adjustments. Following Borusyak et al. (2025), we apply the logarithmic transformation to world export supply prior to interacting with baseline shares, thereby avoiding the mechanical endogeneity that arises from logging the weighted sum.

Identification relies on the assumption that world export supply shocks are orthogonal to unobserved determinants of emission intensity in Swedish manufacturing, conditional on fixed effects and baseline exposure shares. Formally, the exclusion restriction requires that

$$\mathbb{E}\left[IV_{it}^{SAMS} \cdot \varepsilon_{it} \right] = 0,$$

where ε_{it} denotes the structural error in equation (7). Following the shift-share framework of Goldsmith-Pinkham et al. (2020), Borusyak et al. (2022a) and Borusyak et al. (2025), identification can in principle arise from exogeneity of either the shares or the shocks. In our setting, the "shares view" would require that firms' baseline product exposure θ_{ip0} and import shares ϕ_{fp0} are uncorrelated with future unobserved emission trends. Given the well-documented links between product specialization, productivity, and environmental performance, this assumption would be strong (Gullstrand and Knutsson, 2019; Akerman et al., 2024).

[6] Two clarifications on the construction of Z_{spt} are warranted. First, the exposure term $\sum_{f \in \Omega_{s'}} \phi_{fp0}$ aggregates importers by their product-mix composition (the within-firm share ϕ_{fp0} sums to 1 per firm), rather than by baseline import value. This design choice reflects the identification goal: we seek to isolate variation in the *composition* of exposure to global supply shocks, holding constant the importer set. Robustness to size-weighted exposure — replacing ϕ_{fp0} with $\phi_{fp0} \times \bar{M}_{f0}$, where $\bar{M}_{f0}$ is each importer's baseline total import value — yields qualitatively similar first-stage and reduced-form estimates. Second, when a producing firm i is also an importer, its own pre-sample shares ϕ_{ip0} enter the same-SAMS component of $Z_{s(i),pt}$, potentially creating a direct own-import channel for the world supply shock. To verify this does not drive the results, we re-estimate using a leave-one-out version of Z_{spt} that excludes the producing firm's own shares from the same-SAMS term. The baseline estimates are unchanged, confirming that the instrument captures competitive pressure from *other* importers rather than the firm's own exposure.

Instead, our identification relies on the "shifts view." We exploit variation in product-level world export growth occurring outside Sweden and its neighboring markets. These foreign supply shocks reflect changes in technology, trade costs, and productivity in exporting countries. Conditional on firm fixed effects, sector-year fixed effects, and municipality-year fixed effects, such shocks are plausibly unrelated to idiosyncratic emission dynamics of Swedish upstream producers.

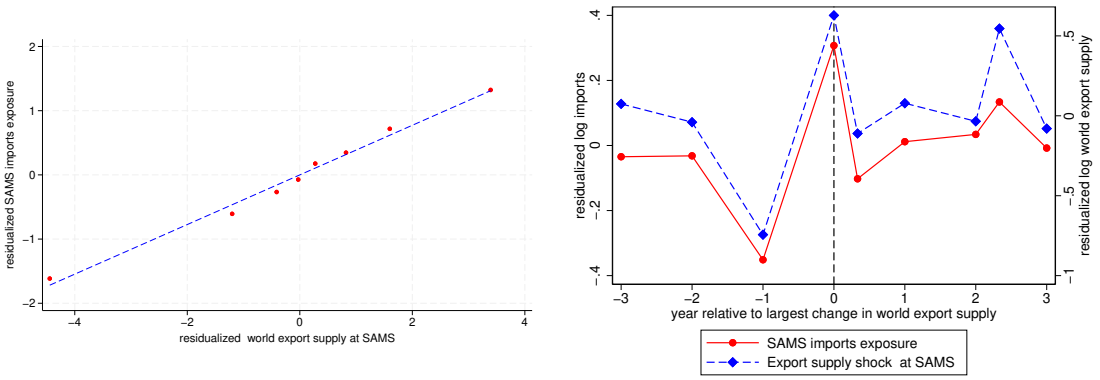
Intuitively, the design compares firms that are differentially exposed to the same global product shock through predetermined import linkages. Because baseline shares are fixed prior to the sample window, and world export growth is measured in third markets, the identifying variation comes from foreign supply shifts rather than domestic demand conditions or contemporaneous firm adjustments. We further assess this assumption by conducting shift-level balance tests and event-study pre-trend analyses, following Borusyak et al. (2025). We also control for firm, industry and location sorting by including firm fixed effects (θ_i), sector-year fixed effects (θ_{bt}) and municipal-year fixed effect. As robustness, we also add producer's own import, information on firm capital, employment, investment in machines, and the amount of emission rights purchased as additional controls.

First stage and reduced-form evidence. Before turning to the structural estimates, we provide graphical evidence on both instrument relevance and reduced-form effects. Figure 2 examines the relationship between the instrument and localized IC. Following Akerman et al. (2024), we residualize both the IC measure and the export shock with respect to firm fixed effects and five-digit industry–year fixed effects, isolating the variation that identifies our specification. The left panel shows a clear and monotonic positive relationship, consistent with the first-stage regression. Across specifications, the Kleibergen–Paap F-statistic exceeds the conventional threshold of 10, indicating strong instrument relevance. The event-study panel in Figure 2 further illustrates that localized IC responds sharply and contemporaneously to large export-supply shocks. The absence of pre-trends supports the interpretation that variation in IC is driven by foreign supply shifts rather than underlying local trends.

Figure 3 presents the corresponding reduced-form relationship between the instrument and producers' log CO_2 emission intensity. The left panel shows a stable negative association, closely mirroring the model specification in Table 1 and indicating that global export shocks translate into cleaner production outcomes. The event-study in the right panel traces emission intensity around the timing of the largest export shock faced by each producer. By construction, the mean log world export supply exhibits a sharp increase at time zero, reflecting the largest shock in the instrument. We find that an increase in export shocks correlates with a nearly immediate decline in firms' emission intensity. Thus, our assumption of contemporaneous effect of local IC and emission intensity seems tenable and appropriate for our model in equation (7).

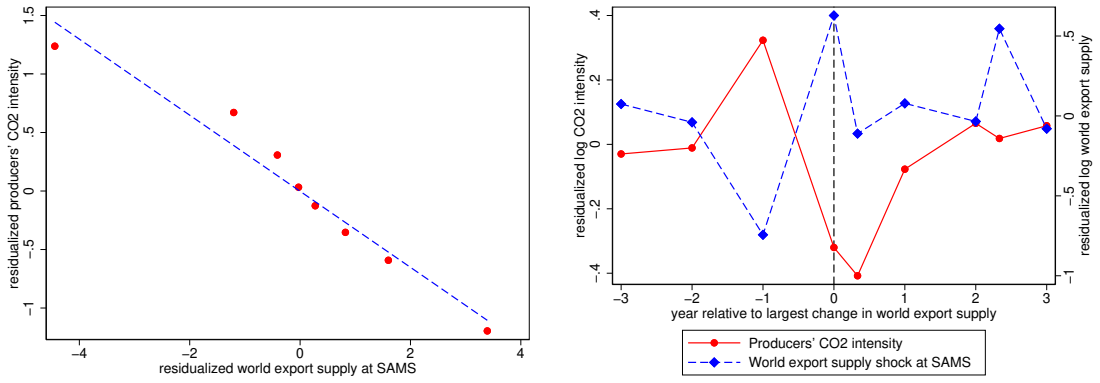
To further corroborate this timing assumption, Figure 4 presents an IV event-study that directly traces the dynamic response of emission intensity to instrument-induced shocks in localized IC. Reassuringly, we observe no statistically significant pre-trends in the years preceding

Figure 2: Export Supply Shock at SAMS level and Local IC



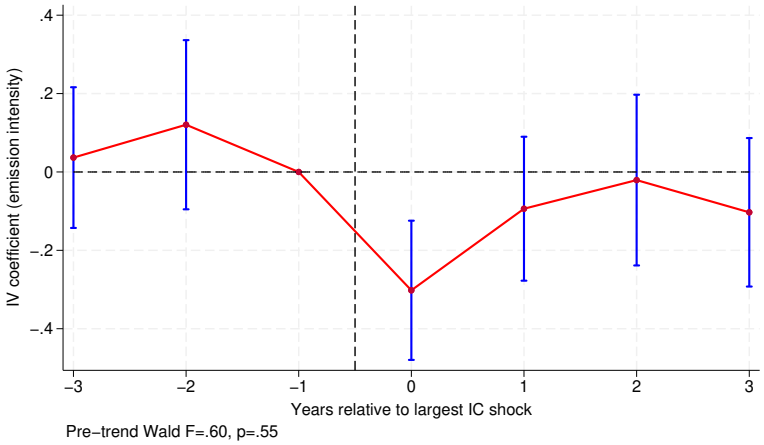
Note: The graph on the left depicts the estimated first-stage relationship between the log of the instrument (IC^{IV}_{sit}) and the log of IC exposure at SAMS level. The variables are residualized by regressing them on firm and five-digit industry–year fixed effects, with the residuals saved for plotting. The data is grouped into eight bins based on the residualized instrument, ensuring each bin contains an equal number of firms. The graph on the right shows the trend of the average log of the instrument and the average log of IC relative to an event-time variable. This event-time variable is set to zero in the year in which each firm experiences its largest increase in the export supply shock at the SAMS level. The analysis is limited to firms with data available for three years before and after the largest observed increase in the instrument.

Figure 3: Export Supply Shock at SAMS level and Producer's CO_2 Intensity



Note: The graph on the left presents the estimated (intention-to-treat) relationship between the log of the instrument (IC^{IV}_{sit}) and the log of CO_2 emission intensity of producers. As with the Figure 2, the two variables are residualized by regressing them on firm and five-digit industry–year fixed effects, with residuals saved for analysis. The firms are then grouped in 8 equal bins based on the residualized instrument. The graph on the right shows the trajectory of the average log of the instrument and the average log of CO_2 emission intensity relative to an event-time variable. This event-time variable is set to zero in the year in which each firm experiences its largest increase in the instrument. The analysis is restricted to firms with available data spanning three years before and after the largest observed increase in the instrument.

Figure 4: Dynamic Effect: Export Supply Shock and Producer's CO_2 Intensity



Note: This figure plots IV event-time coefficients of log CO_2/value added relative to the year in which each firm experiences its largest instrument-induced increase in localized IC (event time 0). The dashed vertical line denotes the event year. Coefficients are estimated relative to $t = -1$. Blue bars represent 95% confidence intervals. The pre-trend Wald test fails to reject joint insignificance of the leads (F=0.60, p=0.55), indicating no evidence of differential pre-trends. Emission intensity declines sharply at the time of the shock, consistent with a contemporaneous competitive response.

the shock. Emission intensity declines sharply in the event year, with the largest adjustment occurring contemporaneously. The effect partially attenuates in subsequent years but remains negative on average. This pattern strengthens the interpretation that competitive pressure operates quickly, consistent with immediate operational and efficiency adjustments rather than gradual pre-existing trends.[7]

5 Results

5.1 Main Results

We begin by investigating the spatial impact of IC defined at the SAMS level on the emission intensity of producing firms. Table 1 shows the results of both the OLS and IV (instrumental variable) estimates for estimating equation (7). In columns (1) and (4), we measure IC without any restrictions on the distance between the SAMS region of the producing firm and importing firms in all other SAMS regions in Sweden. In columns (2) and (5), we divide the IC effect across two distance bands 0-500 km and 500-1000 km from the producing firm. Finally in columns (3) and (6), we measure IC at the national level, i.e. IC is measured as the sum of imports from all firms importing the products that are simultaneously produced by the local

[7]While Figure 4 indicates that the bulk of the adjustment in emission intensity occurs contemporaneously with the IC shock, we do not rule out the possibility of additional gradual responses. Firms may require time to undertake longer-term investments in process innovation, technology adoption, or organizational restructuring. To account for this possibility, we extend our analysis by incorporating lagged measures of IC exposure as a robustness check. The results remain statistically significant and economically meaningful, suggesting that although the primary adjustment is immediate, competitive pressure may also induce persistent or incremental improvements over time.

producing firm within Sweden. Focusing on the IV estimations, the results show that a 10 percent increase in local IC is associated with a statistically significant 5.7% percent decline in emission intensity in the IV estimation in column (4).[8] The coefficient is economically much larger than the analogous national-level IC estimate in column (6). The geographic nature of the effect is further clarified by the distance-decay estimates. As shown in column (5), the IV point estimates attenuate monotonically as we widen the distance bands from 0–500 km to 500–1000 km. This clear decay with distance provides direct evidence that proximity matters: competitive pressures from geographically nearby importers exert the strongest influence on upstream producers' emission intensity. These results confirm that localized measures of IC generate larger estimated reductions in firm-level emissions than national measures, consistent with local proximity amplifying competitive pressure. The consistently strong Kleibergen-Paap (KP $>$ 10) statistics indicate that the instruments are sufficiently strong and relevant for our analysis (as discussed above), and strengthen the causal interpretation of the findings.

To put this in context, Chinese exports to Sweden grew by an average of 21% annually between 2005 and 2014. Assuming Chinese exports, sometimes referred to as Chinese import penetration, are equivalent to IC in Sweden, our results suggest a decrease of around 12% on average in emission intensity in Swedish firms in response to this IC. While not directly comparable, Swedish GHG emission intensity (Kilotons CO_2 eq. per one Swedish Krona) dropped by around 20% between 2008 and 2014.[9]

Overall, the findings indicate that as we widen the geographical scope from the SAMS area demarcations to the national level, the relationship between imports and the firm's emission intensity weakens. This does not mean that national measures are simply "too small": rather, the response is concentrated among the most locally exposed firms, so a national measure—which averages exposure over space—attributes a smaller effect to the average firm and obscures where the supplier adjustment actually occurs. Recovering it requires measuring competition at the scale at which it operates. Our results align with Gullstrand and Knutsson (2019) who found that IC does not significantly affect the firm's marginal costs when measured at the national level, but rather has a significant impact at the local (SAMS) level. This is expected because it is likely that trade shocks will have a more pronounced impact on producers in close proximity, and the effects of these shocks are likely to falter with distance. In terms of production efficiency, the results suggest IC across specific spatial areas will make firms more energy efficient with better production decisions and techniques. We investigate such possible mechanisms in section

[8] A potential concern with our baseline outcome — $\log(CO_2/\text{value added})$ — is that it can fall mechanically if IC raises value added without reducing physical CO_2. The mechanism results (Section 6, Table 5) confirm that IC does raise value added, consistent with productivity gains, so this denominator effect is a live concern. Column (1) of Table 4 addresses this directly by replacing the denominator with physical output; the estimated effect remains negative and statistically significant, suggesting genuine improvements in environmental efficiency. We acknowledge that physical output is measured in heterogeneous units across products (kilograms, liters, units); the consistency of the negative effect across both denominators strengthens confidence that the decline in EI reflects real abatement rather than a mechanical valuation effect.

[9] Source for this information is SCB. No information is available before 2008. Source: SCB Swedish emission intensities.

Table 1: Main Results

	Dep. var: log (CO_2 per value added)					
	(1)	(2)	(3)	(4)	(5)	(6)
IC_{it}^{SAMS}	-0.235***			-0.569***		
	(0.020)			(0.050)		
$IC_{it}^{SAMS^{0-500km}}$		-0.102***			-0.299***	
		(0.017)			(0.061)	
$IC_{it}^{SAMS^{500-1000km}}$		-0.054***			-0.147***	
		(0.013)			(0.047)	
IC_{it}^{NAT}			-0.074***			-0.156***
			(0.006)			(0.012)
Observations	7,924	7,924	7,924	7,924	7,924	7,688
KP				143.93	98.89	214.83
Estimation Strategy	**OLS**	**OLS**	**OLS**	**IV**	**IV**	**IV**
Firm FE	√	√	√	√	√	√
Sector-Year FE	√	√	√	√	√	√
Municipal-Year FE	√	√	√	√	√	√

Note: IC_{it}^{SAMS} is the import competition measure weighted by the inverse distance function. We also use other distance band weights such as as 0-500km ($IC_{it}^{SAMS^{0-500km}}$) and 500-1000 km ($IC_{it}^{SAMS^{500km-1000km}}$). IC_{it}^{NAT} is producer i's IC exposure measured for at the national level in Sweden (i.e. IC from all firms in Sweden importing the same products produced by producer i . The regressions sample contains 1,497 producing firms in Sweden. For each estimation specification, we include firm fixed effects to control for time-invariant unobservable differences across producing firms. We account for sector-year fixed effect to purge the residual from all time-varying industry shocks. We also include municipal-year fixed effects to absorb shocks in municipal-level labor market dynamics. Statistical significance levels are denoted by * for p<0.10, ** for p<0.05, and *** for p<0.01. Standard errors are clustered at both the firm and SAMS levels to allow for arbitrary heteroskedasticity, serial correlation within firms, and spatial correlation within SAMS regions over time.

6. Before that, we test our results for robustness next.

5.2 Robustness and Heterogeneity

We subject our benchmark result in column (5) of Table 1 to a host of robustness checks as follows.

Geographic Composition and Spatial Frictions. The baseline results indicate that localized IC reduces emission intensity. Furthermore, our analysis in section 2.3 suggests that there is geographic concentration of importing activities around the industrial centers of Stockholm and Gothenburg. A natural concern, therefore, is that these estimates may reflect geographic composition rather than true competitive pressures. Specifically, if cleaner industries are disproportionately located near industrial hubs, ports, or coastal cities, the observed IC effect could simply capture differences between coastal and interior regions. To address this concern, we implement a set of robustness checks that account for port proximity, coastal status, importer-side geography, and alternative nearest-neighbor definitions of IC.

Table 2: Robustness I: Geographic Location

	Baseline	Drop Stockholm/Gothenburg	Non-Port Municipality	Port Municipality	Coastal Interaction	Interior Sample	Urban Density
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
IC_{it}^{SAMS}	-0.569***	-0.576***	-0.644***	-0.507***	-0.513***	-0.502***	-0.535***
	(0.050)	(0.047)	(0.073)	(0.061)	(0.076)	(0.105)	(0.083)
$IC_{it}^{SAMS} \times Coastal$					0.012		
					(0.013)		
$IC_{it}^{SAMS} \times Urbanization$							-0.015
							(0.031)
Observations	7,924	7,611	5,245	2,534	4,508	2,828	7,924
Estimation Strategy	**IV**	**IV**	**IV**	**IV**	**IV**	**IV**	**IV**
KP	143.93	152.80	79.70	74.65	24.23	20.95	55.98
Firm FE	√	√	√	√	√	√	√
Sector-Year FE	√	√	√	√	√	√	√
Municipal-Year FE	√	√	√	√	√	√	√

Note: This table reports robustness checks of the baseline specification to alternative geographic samples and interactions. Column (1) reproduces the baseline results. Column (2) excludes firms located in Stockholm and Gothenburg, Sweden's two largest metropolitan areas, to ensure that the results are not driven by urban agglomerations. Column (3) restricts the sample to non-port municipalities, while Column (4) focuses on port municipalities, to assess whether proximity to international trade infrastructure alters the estimates. Column (5) interacts IC_{it}^{SAMS} with an indicator for coastal municipalities to test whether IC has heterogeneous effects depending on coastal access. Column (6) restricts the sample to interior municipalities, and Column (7) interacts IC_{it}^{SAMS} with a measure of urban density to capture differential effects in more urbanized areas. All specifications are estimated using instrumental variables (IV), with the Kleibergen–Paap (KP) statistic reported for instrument strength. Each regression includes firm fixed effects to control for time-invariant heterogeneity, sector-year fixed effects to absorb sector-specific shocks and trends, and municipality-by-year fixed effects to account for local demand or policy shocks at the municipal level. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. Standard errors are clustered at both the firm and SAMS levels to allow for arbitrary heteroskedasticity, serial correlation within firms, and spatial correlation within SAMS regions over time.

We begin by excluding Stockholm and Gothenburg from the sample (Table 2, column 2). The results remain virtually unchanged relative to the baseline, indicating that these hubs do not drive our findings. Next, we compare firms located in port municipalities with those in non-port municipalities (columns 3 and 4 respectively). The estimated effects are similar across both groups, with non-port municipalities exhibiting a slightly larger effect of 6% compared to 5.1 % in port municipalities.[10] This comparison suggests that proximity to seaports does not dominate the IC channel. We then interact IC with proximity to coastal areas, defined as

[10]List of municipalities with ports are sourced from https://www.searates.com/maritime/sweden.

producers within 50 kilometers of a seaport, and find the interaction effect to be small and statistically indistinguishable from zero. This result confirms that the IC effect is not driven exclusively by coastal firms.[11]

We further examine an interior-only sub-sample, defined as firms located more than 50 kilometers from the nearest port, and re-estimate the instrumental variable specification (column 6). The interior-only estimates continue to show a sizable and somewhat similar negative effect. This pattern reinforces that the mechanism underlying IC is not confined to coastal regions. We also interact IC with the degree of urbanization and find no significant effects, suggesting that urban concentration does not drive the results.

Finally, we perform a jackknife analysis across municipalities, sequentially omitting each municipality from the sample. We find that the coefficient estimate is very stable in this exercise, suggesting no single municipality drives the results, further confirming the robustness of our findings (Figure C.1).

Across all of these specifications, the estimated effect of localized IC on emissions intensity remains robust. These results demonstrate that our conclusions are not sensitive to alternative operationalizations of spatial frictions or regional composition, and they provide compelling evidence that the observed effects reflect genuine competitive pressures rather than geographic artifacts.

Nearest Neighbor. One reservation about using the spatial units defined by the SAMS regions may be that their geographical demarcations (administrative borders) are somewhat arbitrary. Hence, an importing firm just within the SAMS border is given a higher weight than a firm just outside the border although proximity to the local producer is the same. This may introduce some bias to the IC measure. Also, recall that we set the internal distance within a SAMS region equal to 1 km, which may underestimate the internal distances within a SAMS area. To overcome the potential bias from the previous measure, we consider alternative definitions of IC based on the x nearest importing firms to the local upstream producer as follows:

$$IC_{it}^{xn} = \sum_{f \in \Omega_{x, f \neq i}} Import_{fjt};$$

where the IC faced by the upstream producer IC_{it}^{xn} is the sum of all imports of product j imported by the importing firms f, which are part of the set of the x nearest importing firms Ω_x. We use various counts of nearest neighbors. To explore how IC behaves as we consider higher x^{th} nearest neighbors, Table 3 shows the coefficient estimates of IC from estimating

[11] We hand-collect all coordinates (latitudes and longitudes) of ports in Sweden from https://www.searates.com/maritime/sweden. We then calculate the nearest distance of each firm's production facility to the ports using STATA's `geonear` command. The resulting variable, *distance to port*, is measured in kilometers and denotes the minimum distance from each producer to the closest port. To capture coastal proximity, we define a binary indicator, *Coastal*, equal to 1 if the facility is located within 50 km of a port, and zero otherwise. This threshold follows common practice in the trade-transport literature (see Faber, 2014 for a similar approach) and is varied in robustness checks (25 km and 75 km) to ensure absence of sensitivity.

equation (7) for the 1st up to the 4th nearest importing firm. We observe that the magnitude of this coefficient drops significantly after the third nearest neighbor. In other words, it seems that IC introduced by potential buyers that are nearest to the producing firms makes the latter more energy efficient and this effect dies out the further out the importing firm is from the upstream producer. Quantitatively, we find that for a distance of about 50 km between importer and producer (an average distance for two-nearest neighbors), a 10% increase in IC can reduce the producers' emission intensity by approximately 2.4%.

Table 3: Robustness II: Nearest Neighbor

	$x = 1$	$x = 2$	$x = 3$	$x = 4$	$x \leq 5$	≤ 10
	(1)	(2)	(3)	(4)	(5)	(6)
IC_{it}^{xn}	-0.536**					
	(0.245)					
IC_{it}^{xn}		-0.243***				
		(0.090)				
IC_{it}^{xn}			-0.289			
			(0.189)			
IC_{it}^{xn}				-0.155*		
				(0.089)		
IC_{it}^{xn}					-0.336**	
					(0.144)	
IC_{it}^{xn}						-0.229**
						(0.097)
Observations	7,559	7,559	7,469	7,391	7,776	7,776
KP	11.57	38.47	8.70	24.90	12.41	16.85
Firm FE	√	√	√	√	√	√
Sector-Year FE	√	√	√	√	√	√
Municipal-Year FE	√	√	√	√	√	√

Note: IC_{it}^{xn} denotes import competition, measured as exposure to the x geographically closest neighbors. All specifications include a rich set of fixed effects: firm fixed effects to control for time-invariant firm characteristics, sector-year fixed effects to capture sector-specific shocks and trends, and municipality-by-year fixed effects to absorb local demand or policy shocks at the municipal level. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. Standard errors are clustered at both the firm and SAMS levels to allow for arbitrary heteroskedasticity, serial correlation within firms, and spatial correlation within SAMS regions over time.

As part of this robustness exercise, we consider different aggregations of the nearest importing neighbors. We first consider the five nearest neighbors ($x \leq 5$), and then the 10 nearest importing neighbors ($x \leq 10$). If a producer has multiple importers that are equally distant, we select the largest of these importers. In most instances, the distances to the nearest neighbors encompass a larger area than the SAMS administrative boundaries, which suggests that the SAMS regions may be somewhat restrictive as spatial units. When we compare the total import value of the five nearest importing manufacturing firms to the total import within a administrative border, the sum of the five closest importers exceeds the import value at the SAMS level in 77% of cases, and in 67% of cases when using municipalities. We present the results of the 5 and 10 nearest importing neighbors in columns (5) and (6) respectively of Table 3. The results show that a 10% increase in IC stemming from the five nearest importers reduces emission intensity by the emitting producer by about 3.4%. This effect decreases to about 2.3% if the 10 nearest neighbors are considered. Hence, these alternative definitions complement our SAMs-based

and distance-band measures and allow us to trace the distance decay of IC effects in a high-resolution setting. Together, these tests directly address the possibility that our findings reflect geographic composition rather than localized competition.

Alternative measure of emission intensity. We investigate the robustness of our benchmark result to changing the dependent variable to emissions per unit of output. The result in column (1) of Table 4 suggests that our benchmark estimate is qualitatively robust to this change, suggesting that firms reduce CO_2 emissions not just per unit of output but also relative to their economic contribution in response to IC.

Balanced Sample and Lagged effect. Another potential concern is that the observed decline in emissions intensity may reflect firm entry and exit in response to IC, which would make the estimates sensitive to changes in firm composition. To address this, column (2) of Table 4 reports results for a balanced panel of firms present throughout the sample period. The estimates remain largely unchanged, indicating that our findings are not driven by sample composition or entry and exit dynamics.

While our main specification captures the immediate response to IC, column (3) of Table 4 allows for lagged effects to account for potential gradual adjustments that firms might undertake, such as process improvements or incremental technology adoption. The coefficient increases modestly, consistent with adjustment over time.

Role of Swedish environmental policy. Sweden has one of the highest CO_2 tax rates in the world and is part of EU emissions trading scheme (ETS). This raises a natural concern: rather than improving productivity, firms facing stronger IC might instead switch to cheaper but dirtier modes of production, with Sweden's environmental policies merely constraining the extent of such behavior. To address this, we implement a series of robustness checks that directly test whether our findings are sensitive to energy prices or Sweden's carbon pricing regime. The results, reported in Table 4 (columns 4-5), show that our baseline effect is highly robust. Column (4) extends the specification to include firm-specific energy prices. We also include additional firm-level controls such as producer's own import, capital, employment, investment in machinery, and ETS allowances purchase. If the effect were explained by a shift to lower-cost, higher-emission fuels, the inclusion of energy costs would substantially reduce the coefficient. Instead, the estimate remains nearly unchanged (–0.568), directly contradicting the "dirty production" hypothesis. Subsequently, we account for Sweden's evolving carbon pricing policies. Following Martinsson et al. (2022), we construct a firm-year measure of the effective marginal CO_2 tax rate, which incorporates statutory tax schedules, sectoral exemptions, revenue-share thresholds (0.8% before 2011, 1.2% thereafter), and EU ETS participation. Column (5) includes this measure as an additional control. The coefficient remains highly stable (–0.568), indicating that our results are not mechanically driven by differential exposure to carbon taxation.

Heterogeneity. We also subject our results to several heterogeneity checks. The coefficient estimates produced from this series of estimations are plotted in Figure D.1. First, we examine whether exposure to IC affects firms differently based on the emission intensity of their production. Specifically, we analyze whether the impact of IC varies depending on the emission

Table 4: Robustness III: +Additional Controls

	Unit Output	Balanced Sample	Lagged Effect	Controls +Energy Price	Controls + Carbon tax
	(1)	(2)	(3)	(4)	(5)
IC_{it}^{SAMS}	-0.819***	-0.563***	-0.565***	-0.568***	-0.568***
	(0.070)	(0.069)	(0.120)	(0.046)	(0.046)
Observations	7,924	2,634	3,887	7,924	7,924
KP	143.93	101.38	16.62	39.45	37.96
Firm FE	√	√	√	√	√
Sector-Year FE	√	√	√	√	√
Municipal-Year FE	√	√	√	√	√

Note: Column (1) reports results where emission intensity is defined as emissions per unit of output. Column (2) restricts the sample to a balanced panel of firms, thereby excluding entry and exit dynamics and ensuring that the estimates are not driven by compositional changes in the firm population. In Column (3), IC is lagged by one period to mitigate potential simultaneity concerns. Columns (3) and (4) further examine the role of energy prices and the Swedish carbon tax in shaping firm-level emissions. Columns (4) and (5) additionally control for firm-level characteristics, including domestic firm's own imports, capital stock, employment, machinery investment, and purchases of emission trading rights (ETS), to capture heterogeneity in production capacity, labor intensity, and compliance behavior. Statistical significance levels are denoted by * for p<0.10, ** for p<0.05, and *** for p<0.01. Standard errors are clustered at the firm and SAMS level to account for heteroskedasticity and within-firm and SAMS correlation over time.

intensity of the firms. To this effect, firms are classified as dirty (clean) if their energy use is above (below) the median values of the sample. The results in panel A show that both clean and dirty firms observe lower emission intensity, but this effect is larger for cleaner firms. This indicates that environmentally proactive firms respond more positively to competitive pressures by further reducing their emissions, possibly through enhanced efficiency and adoption of cleaner technologies as we discuss later.

Second, we examine differences between firms subjected to the EU-ETS and those that are not. We split the sample into two, one for non-ETS firms and another for ETS firms respectively. Estimating the baseline specification with the two samples separately yields a larger coefficient estimate for non-ETS firms (panel B). This suggests that while ETS compliance may reduce emissions, it also narrows the scope for further improvements from non-price stimuli such as IC. This result aligns with studies such as Abrell et al. (2022), who have shown that regulated firms under the EU-ETS often operate closer to their abatement frontier, and additional policy shocks, such as higher carbon prices, yield smaller marginal reductions in CO_2 emission intensity.

Third, we check the heterogeneity of our results along the firm size. We estimate the baseline specification separately for small- and medium-sized enterprises (SMEs) with less than 250 employees on the payroll, and large firms with 250 or more employees. The coefficient estimates (panel C) suggest that while both types are affected similarly by IC, larger firms reduce their emission intensity by around 50% as much following a given increase in IC. This is expected because larger firms tend to be more productive and possess more financial space to respond to shocks. A similar exercise attempts to differentiate between MNEs (multinational enterprises) and non-MNEs (panel D). The coefficient estimate is larger for MNEs and reflects the larger adjustment capacity of MNEs in the face of IC shocks.

Finally, we investigate whether IC coming from countries with less strict environmental regulation has a bearing on the firm's emission response. For example, IC from cheaper and often high emission-intensive products may trigger a different response from the local producer in

Sweden than IC stemming from countries with similar environmental stringency as Sweden. To conduct this exercise, we use the OECD's Environmental Policy Stringency (EPS) index, a composite index derived from indicators of about 15 market-based and non-market environmental policy instruments in 28 countries, including 4 non-OECD countries (Botta and Koźluk, 2014). This indicator ranges from 0 (most lax policies) to 6 (most stringent policies). Using this indicator, we construct a dummy variable that takes the value of one for countries with an average EPS index that is lower than Sweden's score (less strict) between 1991 and 2014, and zero otherwise. The result in panel E indicates that the reduction in emission intensity applies to IC from countries with laxer environmental policies, with the coefficient from the estimation for the sample of countries with stricter environmental code being statistically not different from zero.

6 Channels at Work

While the results demonstrate that IC reduces firms' emission intensity, the precise mechanisms driving these reductions remain an open question. Several potential channels merit further investigation. First, reductions in emission intensity could stem from an overall increase in firm productivity, which leads to fewer inputs (e.g., fuel) per unit of output. Second, firms may respond to competitive pressure by adopting cleaner technologies, investing in energy-efficient production processes, or shifting toward lower-carbon inputs. Third, firms might outsource pollution-intensive activities to third-party suppliers or relocate production to regions with less stringent environmental regulations.

Efficiency and Pro-competitive Effect. First, we assess the impact of localized import exposure on the productivity of surviving firms. Existing research has established that low competition tends to diminish the firm's incentives to streamline its production processes and increase efficiency (Jabbour et al., 2019). Thus, we hypothesize that increased localized IC exposure can lead to a reduction in slack among surviving producing firms and enhance within-firm productivity (Newman et al., 2023). This boost in firm productivity will result in fewer inputs required to produce a given output (Copeland et al., 2022); therefore, the amount of fuel needed will be less. We put this to the test by regressing our first IC measure on the firm's productivity. In Table 5, column 1, using firm total factor productivity (TFP) as measured by Wooldridge (2009), we find a positive and significant impact of IC on the productivity of the upstream producer's productivity; a 10% increase in IC leads to an increase of 0.25% in firm productivity.[12] When comparing this estimate to the estimate produced by Akerman et al. (2024) in their study on the effects of direct importing on the productivity of Swedish manufacturing firms, our estimate is smaller in magnitude. This suggests that although importing firms benefit in terms of productivity gains from an increase in imports, so do non-importing producing firms through IC.

The productivity gains from IC are further reflected in other firm performance metrics, such

[12] Using an alternative method for measuring TFP by Levinsohn and Petrin (2003) yields a similar result.

Table 5: Potential Channels of Transmission

	TFP	VA	MC	P	MU	Core Product	Prob(Product Exit)		Abatement	Offshoring
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
IC_{it}^{SAMS}	0.025*	0.014*	-0.457***	0.208**	0.633***	0.225***	-0.029*	-0.018	0.719**	0.032**
	(0.014)	(0.007)	(0.107)	(0.095)	(0.119)	(0.041)	(0.015)	(0.016)	(0.345)	(0.013)
IC_{it}^{SAMS} × DirtyProdRank								0.003***		
								(0.001)		
Observations	7,721	7,986	7,334	8,028	7,334	8,029	8,029	8,029	879	8,503
KP	63.59	136.47	62.35	71.77	62.35	43.76	43.76	19.68	41.04	63.07
Firm FE	√	√	√	√	√	√	√	√	√	√
Sector-Year FE	√	√	√	√	√	√	√	√	√	√
Municipal-Year FE	√	√	√	√	√	√	√	√	√	√

Note: All regression specifications includes firm fixed effects (Firm FE), Year fixed effects (Year FE), sector-year fixed effects (Sector-Year FE), and municipal-year fixed effects (Municipal-Year FE). TFP stands for Total factor productivity and is estimated using the method by Wooldridge (2009). VA, MC, P, and MU denote the firm's value added, marginal cost, price and markup respectively. We estimate MC and MU based on De Loecker et al. (2016). Firms report abatement investments and these include the installation of pollution control equipment in factories, the development of renewable energy projects, or the implementation of waste management systems. We only have abatement data for a subset of the firms in our sample. Statistical significance levels are denoted by * p<0.10, ** p<0.05, and *** p<0.01. Standard errors are clustered at both the firm and SAMS levels to allow for arbitrary heteroskedasticity, serial correlation within firms, and spatial correlation within SAMS regions over time.

as the firm's value added which increased by about 0.14% in response to an increase of 10% in IC. We further test the hypothesis that the firm's response to IC can lead to *pro-competitive effects* by (i) decreasing their marginal cost, which arises from efficiency gains and reallocating resources to more productive firms, and/or (ii) improving their ability to charge higher prices due to various factors that enable the producing firms to set their products apart, potentially contributing to their productivity improvements (Antoniades, 2015). We estimate markups following closely the estimation approach of De Loecker et al. (2016). For multi-product firms, firm-level price and marginal cost are quantity-weighted averages across the firm's products, with product-level prices recovered as unit values (deflated revenue per physical unit) from the IVP register; the firm markup is the output-elasticity-to-revenue-share ratio aggregated to the firm level.[13] Our findings, presented in columns (3)–(5) of Table 5, indicate that a 10% increase in exposure to IC leads to a decrease of about 4.6% in the firm's marginal cost and an increase of 2.1% in prices. These two effects – lower marginal costs and higher prices – combine into a 6.3% increase in markups for a 10% rise in IC. At first glance this is puzzling: pro-competitive pressure is usually expected to compress markups, not raise them. As set out in the conceptual framework (Section 3), that intuition holds only conditional on a fixed product and fixed quality. It does not describe a supplier that responds to competition by *escaping* it (Aghion et al., 2005): upgrading quality and reallocating toward its more differentiated, higher-margin varieties (Melitz and Ottaviano, 2008; Antoniades, 2015). Differentiation raises both price and markup precisely because the firm is no longer selling the same good into the same contested market. The rising markup is therefore a signature of escape through differentiation rather than of weakened competition, and it is the same adjustment that lowers emission intensity through

[13]Markups are entered as log markup in all regressions. See Appendix A on how markup is recovered using production based model developed by De Loecker and Van Biesebroeck (2016). To address the well-documented sensitivity of ratio-based markup estimators to extreme input-revenue shares, we winsorize markups at the 1st and 99th percentiles prior to estimation.

the composition margin examined next.[14]

Product-Mix Effect. Furthermore, and as mentioned earlier, there is the possibility that, beyond simply streamlining operations, domestic producers may change their product mix. Namely, domestic producers may increase production of their core products (Gullstrand and Knutsson, 2019), or drop their low-quality (and often more emission-intensive) products (Antoniades, 2015) as a strategic response to compete with imported goods.[15] We explore these mechanisms by considering how localized IC impacts (a) the production of core products, and (b) the probability of dropping dirty products. Table 5 column (6) shows that a 10% increase in IC exposure leads to approximately an 2.3% increase in the firm's core product output. However, when we condition exposure on more emission-intensive products using product-survival analysis, we find that firms tend to phase out their energy-intensive products. In columns (7), we run a regression where the dependent variable is the probability that a product is no longer produced (product exit), and in column (8), we interact the IC variable with a rank indicator where the products are sorted into quartiles in terms of their energy use; a rank of zero is given to the products in the bottom quartile, and a rank of 3 is given to the products in the top quartile (i.e. the dirtiest products). We estimate that a 10% increase in IC raises the probability of product exit by 0.29 percentage points — i.e. a product that faces stronger localized competition is more likely to be dropped from the firm's portfolio. The column (8) interaction clarifies that this pro-exit effect is concentrated among emission-intensive products: the positive coefficient on $IC \times$ DirtyProdRank implies that the exit probability increases more steeply for higher-ranked (dirtier) products, consistent with firms selectively shedding their most polluting product lines. This aligns with the trade literature on the composition effect of trade, where multi-product firms can adjust their product mix towards low-energy-intensive and low-marginal-cost products (Shapiro, 2021; Copeland et al., 2022; Leisner et al., 2023).

Abatement vs Offshoring Effect. Another interesting mechanism for emission reduction at the firm level is whether IC induces producing firms to increase investment in pollution abatement and/or are just offshoring dirty part of their production abroad. We analyze annual abatement data from SCB to investigate this mechanism. Importantly, we gather information on the firms' investment in abatement technology. Examples of pollution abatement investments include the installation of pollution control equipment in factories, the development of renewable energy projects, or the implementation of waste management systems. The regression results

[14]The framework also reconciles why markups rise sharply while measured (revenue-based) TFP rises only modestly: when firms upgrade quality and raise prices, part of the efficiency and quality gain is absorbed into the price index rather than into measured physical productivity, while the De Loecker et al. (2016) markup captures the full price-marginal-cost wedge. We probe this directly using value added per worker, a price-inclusive productivity measure, and by testing whether the price response is concentrated where product mix can adjust.

[15]Peripheral, low-quality product lines are typically produced with older, less specialized, and more energy-intensive processes, so shedding them lowers average emission intensity through composition; Barrows and Ollivier (2018) show that within-firm product mix is a quantitatively important determinant of emission intensity. In our setting this link is not assumed but measured: the dirty-product rank used in column (8) is constructed directly from each product's share in the firm's energy use, so the products firms drop under competition are, by construction, the more emission-intensive ones.

for this mechanism are presented in Table 5 column (9). We observe that IC has a significant and positive effect on abatement investment. We interpret this cautiously: the reported expenditures are largely end-of-pipe controls for conventional pollutants, and because CO_2 has no comparable end-of-pipe abatement technology, the relevant technique channel for carbon operates through energy efficiency and fuel switching rather than through pollution-control capital per se. The abatement result is therefore best read as corroborating evidence that firms facing competition raise environmental effort, complementing the productivity and energy-efficiency gains documented above, rather than as a stand-alone CO_2 abatement channel.

It is also conceivable that producing firms facing IC offshore may offshore some of their production as a means to reduce production costs and regain competitiveness. This would in turn reduce emission intensity at home, especially if the firm offshores its relatively more emission-intensive inputs. In other words, IC can lead to the composition of production to shift towards cleaner production by driving an increase in imports of dirty goods that are not produced in-house. To investigate this channel, we follow the study by Dussaux et al. (2023) to calculate the emissions embodied in imports. To account for the carbon content of import, we first compute the emission intensity of each sector in the source countries using data on total emissions and production from the EORA Global Supply Chain Database (EGSCD). We then calculate the emissions embodied in imported inputs as the sum of emissions weight by the firm's imports. That is,

$$CarbonImport_{it} = \sum_{p} \sum_{c} M_{ict,p\in j} EI_{ct,p\in j}$$

where $EI_{ct,p\in j,}$ is the total emission intensity (i.e. tons of CO_2 per unit of real output) of sector j in sourcing country c at time t, while $M_{ict,p\in j}$ is the real import value of firm i for products p belonging to sector j from sourcing country c, deflated to constant prices using product-level import price deflators from SCB and expressed in the same real-output units as EI.[16] In this exercise, we focus on imported intermediate goods, classified at the 3-digit level according to the 5th revision of the Classification by Broad Economic Categories (BEC). The estimated coefficient of this exercise, reported in Table 5 column (10), suggests that IC leads to an increase in offshoring activities with an elasticity of 0.032. We acknowledge that an increase in the absolute level of *CarbonImport* could partly reflect a scale effect: since IC also raises firms' value added (Table 5, column 2), total input demand — including imported carbon-intensive intermediates — may rise mechanically with output expansion. To partially address this concern, we verify that the elasticity of CarbonImport *relative to domestic output* also rises with IC, consistent with a compositional shift toward imported inputs rather than a pure scale effect; this ratio-based check yields a similar positive and statistically significant coefficient. Hence, both abatement and offshoring seem to be at play in the firm's toolbox to respond to increasing IC pressure.[17]

[16] To ensure dimensional consistency, CN8-level import values are deflated using annual product-level price indices and then mapped to ISIC sectors to align with the EORA MRIO real-output basis. The resulting units of $M_{ict,p\in j}$ are millions of constant-price SEK, directly comparable to the real-output denominator in $EI_{ct,p\in j}$. The product $M_{ict,p\in j} \times EI_{ct,p\in j}$ therefore yields tons of CO_2 equivalent embodied in the firm's intermediate imports from country c.

[17] The finding that both the technique effect and the composition effect are at play contributes to the

Conclusively, our study suggests that efficiency gains via IC extend to firms' environmental behavior. Trade exposure at different locations can enhance environmental efficiency via increased productivity growth, product up-scaling and differentiation, and adoption of sustainable production practices, thereby contributing to a greener and more responsible global supply chain.

7 Conclusion

The classical examination of the impact of IC on various firm outcomes, including environmental performance, typically considers IC at the national level, assuming that competition is symmetric nationwide. Using detailed geographical information about the locations of all manufacturing firms in Sweden during the period 2005–2014, we explore the effects of IC in a spatial framework on the CO_2 emissions of local producing firms. Considering different spatial definitions to measure IC, we find evidence that IC from local importers has a significant negative impact on the emission intensity of domestic upstream producers. However, this effect seems to wane as the distance between the producers and importers increases. We also find evidence of improved productivity, greater environmental and energy-efficiency effort, product differentiation and quality upgrading, and the reduction of energy-intensive products among producing firms in response to localized IC.

In summary, our findings reveal a positive globalization impact on the environment coming from increased IC. Consequently, our results underscore the possibility that prior empirical studies might have undervalued the environmental and productivity gains derived from trade. This lends empirical support to the theoretical arguments presented by Autor et al. (2013), emphasizing the role of local economy in playing a significant part of identifying trade-induced changes in the economy.

By extension, the study highlights that trade's effect on the environment could be positive. This does not, however, consider the possibility of carbon leakages. In fact, IC often entails carbon leakage as we have alluded to in this paper because firms may be outsourcing some of their intermediate input to foreign agents that are dirtier than their Swedish counterparts. This means that while our study suggests that IC may have positive effects on the environment in Sweden, we cannot say that the total effect is positive on aggregate after accounting for carbon leakages. To deal with carbon leakages, the EU is planning to introduce the Carbon Border Adjustment Mechanisms (CBAM) in 2026. This involves levying taxes on carbon-intensive imports to mitigate the carbon offshoring effect. Additionally, the creation of a certification system is recommended to ensure that products meet specific eco-friendly criteria, fostering cleaner production methods on a global scale. Another policy implication of our study is the need for

ongoing debate on firms' emissions decomposition. While studies such as Shapiro and Walker (2018) emphasize the dominance of the composition effect, others — including Dussaux et al. (2023), Akerman et al. (2024), Ustyuzhanina (2022), and Leisner et al. (2023) - highlight the primacy of the technique effect. Our results highlight a novel mechanism: the competitive pressure from imports induces local firms to adopt strategies that combine the two effects.

strategic investment in green technologies. Directing local resources towards R&D and sustainable production methods can foster innovation in green technologies and stands to benefit both domestic and imported goods, contributing to more sustainable development.

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Appendix

A Productivity and Markup Estimates

Firm productivity constitutes one of the primary channels through which import competition may affect firms' performance and environmental outcomes. To quantify this mechanism, we first estimate firm-level total factor productivity (TFP) using the control-function estimator of Wooldridge (2009), which explicitly accounts for the simultaneity between firms' input choices and unobserved productivity shocks.

Assume that firms produce according to a Cobb–Douglas production technology,

$$Q_{it} = A_{it} L_{it}^{\beta_l} K_{it}^{\beta_k} M_{it}^{\beta_m}, \tag{8}$$

where Q_{it} denotes output, L_{it} labour input, K_{it} capital, M_{it} intermediate materials, and A_{it} is Hicks-neutral productivity. Taking logarithms yields

$$y_{it} = \beta_l l_{it} + \beta_k k_{it} + \beta_m m_{it} + \omega_{it} + \varepsilon_{it}, \tag{9}$$

where lower-case variables denote logarithms, ω_{it} is firm-specific productivity observed by the firm but not by the econometrician, and ε_{it} is an idiosyncratic productivity shock. Estimating this production function using ordinary least squares would be inconsistent because firms adjust their input choices in response to productivity shocks. Following Wooldridge (2009), we use intermediate inputs as a proxy for the unobserved productivity component and estimate the production function using a one-step generalized method of moments (GMM) estimator. The approach assumes that productivity follows a first-order Markov process, allowing lagged productivity to summarize firms' dynamic optimization while correcting for the endogeneity of variable inputs. Compared with earlier control-function estimators, the Wooldridge estimator jointly estimates all parameters in a single step, improving both efficiency and inference.

Firm-level total factor productivity is then recovered as the residual from the estimated production function,

$$\hat{\omega}_{it} = y_{it} - \hat{\beta}_l l_{it} - \hat{\beta}_k k_{it} - \hat{\beta}_m m_{it}. \tag{10}$$

The resulting TFP estimates are subsequently used to examine whether import competition affects firms through changes in productive efficiency.

A.1 Recovering Markups and Marginal Costs

To investigate whether import competition also affects firms' pricing behaviour, we recover firm-level markups and marginal costs using the production-based methodology proposed by De Loecker et al. (2016). This approach combines estimated production-function elasticities with firms' expenditure shares and does not require specifying a demand system or assumptions

regarding market conduct.

Suppose product j produced by firm i in period t is generated according to

$$Q_{ijt} = F\left(V_{ijt}, K_{ijt}\right) \exp\left(\omega_{it} + \varepsilon_{ijt}\right), \tag{11}$$

where V_{ijt} denotes flexible inputs, K_{ijt} denotes predetermined inputs, ω_{it} is firm-level productivity, and ε_{ijt} captures transitory productivity shocks or measurement error.

Under cost minimization, the firm's first-order condition for a flexible input implies that the markup is identified as the ratio of the output elasticity of that input to its expenditure share. Product-level markups are therefore given by

$$\mu_{ijt} = \frac{\theta^{v}_{ijt}}{\alpha^{v}_{ijt}}, \tag{12}$$

where θ^{v}_{ijt} denotes the output elasticity of flexible input v, and α^{v}_{ijt} is the expenditure share of that input in product revenue.

An important empirical consideration with this ratio-based estimator is its sensitivity to observations with very small input expenditure shares. Because the expenditure share appears in the denominator of the markup expression, small measurement or estimation errors can generate implausibly large markup estimates when α^{v}_{ijt} approaches zero. This concern is particularly relevant in our application because materials are used as the flexible input and product-level material expenditure shares for multi-product firms are recovered through the input-allocation procedure of De Loecker et al. (2016). To ensure that the estimated markups are not driven by a small number of observations with unusually low expenditure shares, we examine the distribution of both material expenditure shares and recovered markups and assess the robustness of the results to alternative trimming and winsorization thresholds.[18]

Marginal costs are then recovered as

$$mc_{ijt} = \frac{P_{ijt}}{\mu_{ijt}}, \tag{13}$$

where P_{ijt} denotes the observed product price. Following De Loecker et al. (2016), we treat materials as the flexible input when constructing markups. Output elasticities are obtained from the estimated production function, while expenditure shares are computed directly from the data. Because the sample contains both single-product and multi-product firms, we adopt the input-allocation procedure proposed by De Loecker et al. (2016). Production-function parameters are first estimated using single-product firms, for which input allocations are directly observed. These estimates are subsequently used to recover the allocation of firm-level material expenditures across products for multi-product firms, allowing consistent estimation of product-level expenditure shares, markups, and marginal costs. Finally, all product-level esti-

[18] In the baseline analysis, we trim observations with material expenditure shares below the first percentile and above the ninety-ninth percentile. The results are robust to alternative treatments of the tails of the distribution.

mates are aggregated to the firm level and used in the empirical analysis.

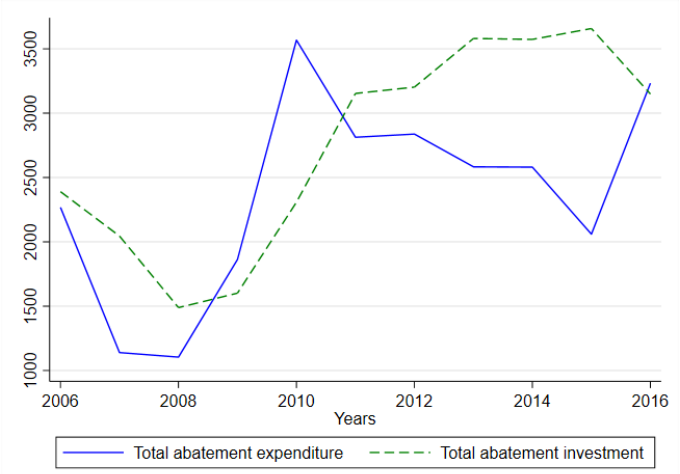
B Descriptive Statistics and Miscellaneous Figures

Table B.1: Descriptive statistics on emissions

	Mean	SD	Min	10th perc.	**Median**	75th perc.	Max
CO_2(1000 tons)	45	189.05	0.003	0.39	2.73	10.05	2378.67
CO_2(tons)/value added (SEK)	0.00017	0.0077	9.11e-07	0.0001	0.0005	0.0012	0.563

Note: The sample comprises 1,497 producing firms with more than 20 employees, observed over the period 2005–2014, which constitutes the regression sample used in the analysis.

Figure B.1: Trend of abatement expenditure and investment (SEK, thousands)



Note: The sample consists of large firms with more than 50 employees. Abatement investments are typically long-term in nature and often require significant capital expenditures upfront. Examples of abatement investments include the installation of pollution control equipment in factories, the development of renewable energy projects, or the implementation of waste management systems. Abatement expenditure, on the other hand, refers to the actual costs or expenses incurred in carrying out abatement measures. It represents the financial outlays associated with implementing and maintaining environmental protection initiatives. Abatement expenditures can include a wide range of costs, such as operational expenses, maintenance costs, monitoring and compliance costs, research and development expenses, or fees paid for environmental permits or licenses.

Table B.2: Descriptive statistics for production, and firm characteristics

	Full sample				Regression sample			
	Mean	SD	Min	Max	Mean	SD	Min	Max
Number of products	9.39	12.84	1	68	2.80	4.30	1	68
Product share in firms revenue	0.28	0.36	0	1	0.53	0.41	0	1
Core product	0.35	0.48	0	1	0.59	0.49	0	1
Product share in firm production$^{t=0}$	0.37	0.39	0	1	0.67	0.40	0	1
Markup	15.9	938.94	0	98092.45	17.31	1148.65	0	98092.45
Employment	547.98	1737.33	20	20492	230.12	931.50	20	20492
Output (various units)/1000	399.78	7259.41	0	450874.09	724.13	11403.84	0	450874.09
Sales (million SEK)	2407.36	9006.02	0	120555.02	930.08	5274.52	0	120555.02

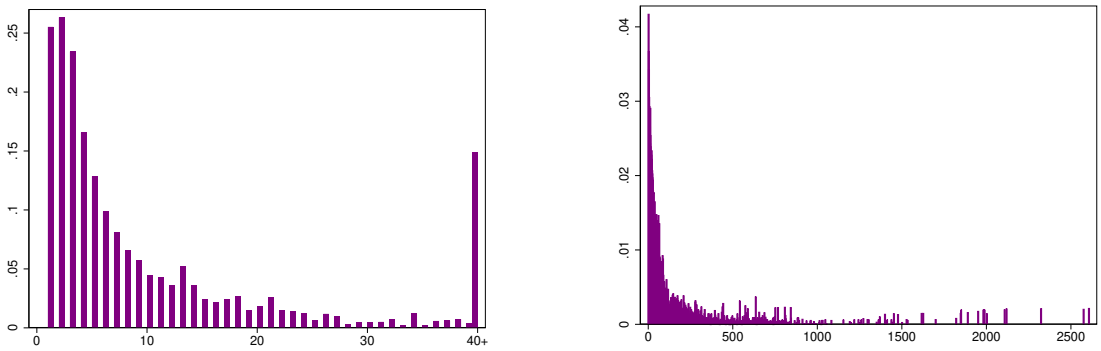
Note: Summary statistics for producing firms in terms of production, size, markup and sales. The left columns summarize the full sample set at firm-plant-product-year level in the database, and the right columns only at the firm-year level with our data restrictions. Our full sample contains 2,279 firms, whereas our regression sample contains 1,497 firms. All observations have a year dimension. Sales and employment are calculated at firm level using the FEK database. Core products are products with shares in firms revenue exceeding 50%. Currency values are millions of current year SEK

Table B.3: Descriptive statistics: Imports and geography

	Regression sample			
	Mean	SD	Min	Max
Import (million SEK):				
Total	190.50	1244.15	0.00	36681.45
SAMS	9.01	99.68	0	7076.78
1st nearest Neighbor	18.31	229.43	0.00	9517.50
2nd nearest Neighbor	17.10	222.86	0.00	9143.51
3rd nearest Neighbor	12.40	149.71	0.00	8738.09
Distance (KM):				
To 1st nearest Neighbor	34.33	63.71	0.00	1027.16
To 2nd nearest Neighbor	52.89	82.99	0.00	1157.26
To 3rd nearest Neighbor	66.32	92.80	0.00	1296.15
Import Competition				
IC^{SAMS}	10.8	2.44	0	17.48

Note: Summary statistics for producing firms in terms of imports (million SEK) and distance (km) from the producer to nearest importing firm . A distance of zero means the nearest neighbor is in the same SAMS area as the producing firm. Import competition measure is in logs. All observations have a time dimension.

Figure B.2: Distribution of Number of producers and importers



A. Number of Producers

B. Number of Importers

Note: The graph shows the distribution of the number of producers (panel A) and importers (panel B) within the SAMS regions of Sweden.

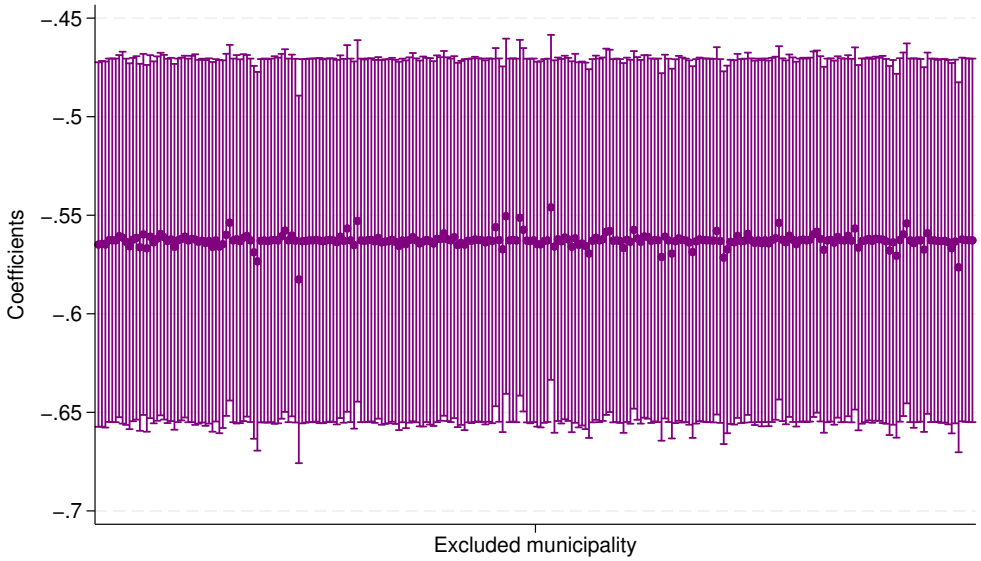
C Robustness Checks

Table C.1: Robustness: Add all distance bands

	Dep. var: log (CO_2 per value added)	
	(1)	(2)
IC_{it}^{SAMS}	-0.221***	-0.479***
	(0.028)	(0.094)
IC_{it}^{NAT}	-0.061***	-0.119***
	(0.005)	(0.014)
$IC_{it}^{SAMS^{0-500km}}$	0.011	0.090
	(0.017)	(0.064)
$IC_{it}^{SAMS^{500-1000km}}$	0.026*	0.107**
	(0.014)	(0.050)
Observations	7,924	7,924
Estimation Strategy	**OLS**	**IV**
KP		20.04
Firm FE	Yes	Yes
Sector-Year FE	Yes	Yes
Municipal-Year FE	Yes	Yes

Note: This table jointly includes the baseline SAMS-weighted IC measure (IC_{it}^{SAMS}), the 0–500 km band measure ($IC_{it}^{SAMS,0\text{-}500\text{km}}$), and the 500–1000 km band measure ($IC_{it}^{SAMS,500\text{-}1000\text{km}}$) in the same regression. Column (1) reports results where emission intensity is defined as emissions per unit of value added. Statistical significance levels are denoted by * for p<0.10, ** for p<0.05, and *** for p<0.01. Standard errors are clustered at the firm and SAMS level to account for heteroskedasticity and within-firm and SAMS correlation over time. **Interpreting sign reversals:** When the baseline and band-restricted IC measures are included jointly, the coefficients represent *conditional* marginal effects — the contribution of each spatial band given the other bands. Because all three measures share common underlying import flows, their simultaneous inclusion induces multicollinearity: the 0–500 km and 500–1000 km coefficients can change sign relative to their stand-alone estimates (reported in Table 1). These sign reversals reflect collinear regressors rather than a genuine violation of monotonic distance decay. The separate-band specifications in Table 1 — each estimated independently — remain the preferred evidence for the decay pattern.

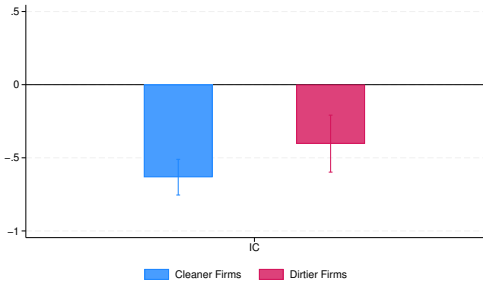
Figure C.1: Jackknife analysis by leaving out one municipality at a time.



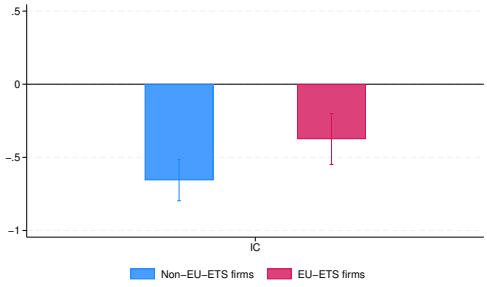
Note: Each point plots the coefficient estimate obtained by re-estimating the baseline specification after omitting the corresponding municipality from the sample (there are 254 municipalities). Vertical lines denote 95% confidence intervals based on two-way clustered standard errors at the SAMS and firm level.

D Heterogeneity

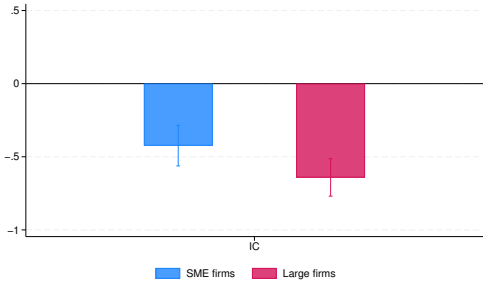
Figure D.1: **Heterogeneity:** IC effects on emission intensity



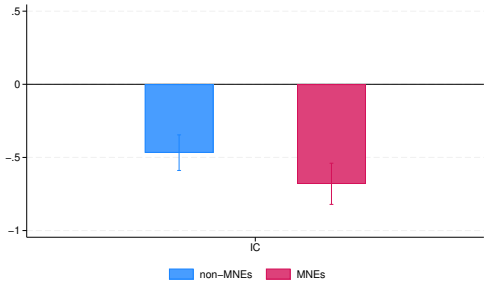
A. Clean vs Dirty firms



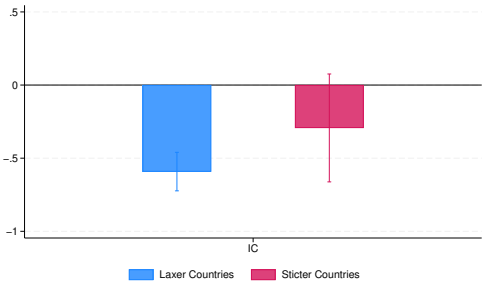
B. EU-ETS vs non-EU-ETS firms



C. SMEs vs Large Firms



D. MNEs vs non-MNEs



E. IC from Laxer vs Stricter Countries

Note: The Figure reports the estimated coefficients β_0 for the different groups defined in equation (7). The vertical axis displays the point estimates, while the bars represent the associated 95 percent confidence intervals. The coefficients are obtained from instrumental variables regressions, where local import competition is instrumented using export supply shocks weighted by pre-sample import shares and the sectoral energy mix. Standard errors are clustered at the SAMS level. In all cases, the Kleibergen–Paap statistics indicate that the instruments satisfy the conventional thresholds for relevance.